It is recommended that you spend about 30 minutes on this question.

2 A simple air-spaced capacitor can be made by placing two metal plates close to each other. When the capacitor is charged the plates have equal and opposite charges as shown in Fig. 2.1.

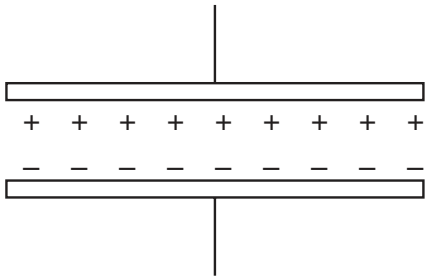

Fig. 2.1

The plates attract each other because they have opposite charges. This force of attraction can be small and therefore difficult to measure.

Design a laboratory experiment to investigate how the force of attraction between the plates depends on the potential difference between the plates. You should draw a detailed labelled diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) how the plates would be charged,

(c) how the potential difference between the plates would be measured,

(d) how the equipment would be arranged so that the force between the plates is made as large as possible,

(e) how the force of attraction between the plates would be measured,

(f) any safety precautions that you would take.

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